DynSGS setups of the paper's numerical tests

decks of branch `sm/MHD` at commit `66ebb9c`; notation of the paper, (11)–(16) and Table *split*

	Smooth vortex, MHD	**Smooth vortex, Euler**[b]	**Brio–Wu**[a]	**Orszag–Tang**	**Rotor**	**Blast**	**Jet, Mach 800**	**Flux emergence**[c]
deck in `problems/MHD/` *unless marked*	`smoothVortex`	`CompEuler/ smoothVortex`	`brioWu1d`	`orszagTang-Bormanis2024`	`rotorDaoNazarov2022`	`blastBalsara-Spicer1999`	`astroJetWuShu2018`	`fluxEmergence-Son2025DSGS`
Discretization (deck; paper, where its text or captions say otherwise)								
Dimension, equations	2D, 9	2D, 4	1D, 8 (no ψ)	2D, 9	2D, 9	2D, 9	2D, 9	2D, 9
γ (`:dsgs_gamma`)	5/3	1.4	2	5/3	1.4	1.4	1.4	1.05
N	4 (figure: a sweep)[d]	4 (figure: a sweep)[d]	4; paper 4, 5, 6, 7	4	3; paper 4	3; paper 4	4	4
Elements	16^2 on $[-10,10]^2$	16^2 on $[-10,10]^2$	$600/N$: 150, 120, 100, 86	64^2 (32^2, refined once); paper 128^2 to 512^2	128^2 (32^2, refined twice); paper 256^2	64^2 (32^2, refined once); paper 128^2, 256^2	80×240 (20×60, refined twice), half domain	160×70 (80×35, refined once)
Δt	$2 \cdot 10^{-3} \cdot 64/(n_{el}N)$	$2 \cdot 10^{-3} \cdot 64/(n_{el}N)$	$5 \cdot 10^{-5} \cdot 600/\mathrm{DOFs}$	$2 \cdot 10^{-4}$; paper 10^{-4} to $2.5 \cdot 10^{-5}$	$2 \cdot 10^{-4}$	10^{-5}	$1.25 \cdot 10^{-7}$	$5 \cdot 10^{-3}$
t_{end}	1	1	0.1	1	0.15	0.01	$2 \cdot 10^{-3}$	50; paper figure 51 τ_0
Time integrator	`CarpenterKennedy2N54` at fixed Δt in every deck, as in the paper							
The coefficient: (11)–(16)								
Sensor R_i (`:dsgs_sensor`)	(R) residual	(R) residual	(R) residual	(L) legacy	(L) legacy	(L) legacy	(R) residual	(L) legacy
$\mu\|_e$ computed (`:dsgs_freeze_stage`)	once per step	once per step	once per step	every RK stage	every RK stage	every RK stage	once per step	every RK stage
C_1, C_2 (`:dsgs_CR`, `:dsgs_Cmax`)	1, 0.5	1, 0.5	1, 0.5[a]	1, 0.5	1, 0.5	1, 0.5	1, 0.5	1, 0.5
Floor $C_{\min}\Delta_e(\|\boldsymbol{u}\| + c_f)$ (`:dsgs_Cmin`)	0	0	0.06[a]	0	0	0	0	0.03
Length Δ_e, (15)	$h_e/(N+1)$	$h_e/(N+1)$	h_e/N[a]	$h_e/(N+1)$	$h_e/(N+1)$	$h_e/(N+1)$	$h_e/(N+1)$	$h_e/(N+1)$
Cutoff c (`:dsgs_cutoff`)	0 in every deck (no cutoff; the Alfvén wave, not in the paper, runs 10^{-3})							
$\mu\|_e = 0$ for the first (steps) (`:dsgs_hold_steps`)	2	2	2	2	2	0	0	2
Residual set $\mathcal{I}$	8, ψ excluded	ρ, $\|\boldsymbol{m}\|$, E	all 8	8, ψ excluded	8, ψ excluded	8, ψ excluded, $+\rho e$ (`:dsgs_thermal_residual`)	8, ψ excluded	8, ψ excluded
Denominator of (12) (`:dsgs_norms`)	domain, Ω	domain, Ω	domain, Ω	domain, Ω	element, Ω_e	element, Ω_e	element, Ω_e	element, Ω_e
Floors σ_i, (16) (`:dsgs_rel`, `:dsgs_local_rel`, `:dsgs_fast_floors`)	$\epsilon_{\mathrm{rel}} = 1$: $\bar{\rho}$, $\bar{\rho}\bar{c}$, $\bar{\rho}\bar{c}^2$, $\sqrt{\bar{\rho}}\,\bar{c}$	$\epsilon_{\mathrm{rel}} = 1$: $\bar{\rho}$, $\bar{\rho}\bar{c}$, $\bar{\rho}\bar{c}^2$	$\epsilon_{\mathrm{rel}} = 1$, as SV	$\epsilon_{\mathrm{rel}} = 1$, as SV	element scales ρ_e, $\rho_e c_e$, $\rho_e c_e^2$, $\sqrt{\rho_e}c_e$, $\times 1$	element scales, $c_e \to c_{f,e}$	element scales, $c_e \to c_{f,e}$	element scales, $\times 1$
Speed in the cap (13)	$\|\boldsymbol{u}\| + c_f^{\max}$	$\rho\,(\|\boldsymbol{u}\| + c)$, dynamic[b]	$\|\boldsymbol{u}\| + c_f$, exact x-fast speed	$\|\boldsymbol{u}\| + c_f^{\max}$	$\|\boldsymbol{u}\| + c_f^{\max}$	$\|\boldsymbol{u}\| + c_f^{\max}$	$\|\boldsymbol{u}\| + c_f^{\max}$	$\|\boldsymbol{u}\| + c_f^{\max}$
Distribution over the equations: the paper's Table *split* (coefficients of ∇ of the diffused variable; $\nu = \mu\|_e$)								
Diffused variables (`:dsgs_conserved`, `:dsgs_ref_weight`)	physical: $\rho, \boldsymbol{u}, T, \boldsymbol{B}, \psi$	physical: $\rho, \boldsymbol{u}, T$	conserved $\boldsymbol{q}$	physical: $\rho, \boldsymbol{u}, T, \boldsymbol{B}, \psi$	conserved $\boldsymbol{q}$	conserved $\boldsymbol{q}$	conserved $\boldsymbol{q}$	$(\boldsymbol{q} - \boldsymbol{q}_e)/\rho_e$; $\boldsymbol{B}, \psi$: $\boldsymbol{q} - \boldsymbol{q}_e$
λ_i (`:μ`)	1 for every equation in every deck, so $\lambda_1 = 1$: mass diffusion is on (the paper: $\lambda_1 = 0$)							
ρ	ν	ν ($\mu/\|\rho\|_{\infty,e}$)	ν	ν	ν	ν	ν	$\rho_e\nu$
$\rho u, \rho v, \rho w$	$\bar{\rho}_e\nu$, stress $\boldsymbol{\tau}$	μ, stress $\boldsymbol{\tau}$	ν	$\bar{\rho}_e\nu$, stress $\boldsymbol{\tau}$	ν	ν	ν	$\rho_e\nu$
E (`:dsgs_nazarov_energy`)	$\bar{\rho}_e\nu/\mathrm{Pr}_t$ on ∇T; $+\boldsymbol{\tau}\cdot\boldsymbol{u}$, $+\eta\boldsymbol{B}\cdot\nabla\boldsymbol{B}$	$\mathrm{P}\,\mu/(\gamma-1)$ on ∇T; $+\boldsymbol{\tau}\cdot\boldsymbol{u}$	ν on E	$\bar{\rho}_e\nu/\mathrm{Pr}_t$ on ∇T; $+\boldsymbol{\tau}\cdot\boldsymbol{u}$, $+\eta\boldsymbol{B}\cdot\nabla\boldsymbol{B}$	ν on E	ν on E	ν on E	thermal part: $\rho_e \max(c_E\nu_c, \nu_{\mathrm{fl}})$, $c_E = \gamma(\gamma-1)/\mathrm{Pr}_t$; rest: $\rho_e\nu$
Pr_t (`:dsgs_Prt`)	1	P = 0.1 (`:Pr`)[b]	1 (not used)	1	1 (not used)	1 (not used)	1 (not used)	1
B_x, B_y, B_z; ψ	$\eta = \nu$; ν	—	ν; —	$\eta = \nu$; ν	ν; ν	ν; ν	ν; ν	ν; ν
Positivity (not part of DynSGS)								
State limiter (`:positivity_method`)	none	node repair; 10^{-6} of the core ρ, p	none	none	conservative; $\rho, p \geq 10^{-6}$	conservative; 10^{-6}, 10^{-7}	conservative; $1.4 \cdot 10^{-7}$, 10^{-6}	none

Red: differs from the paper's text. (R), (L): the residual and legacy sensors of `dsgsCutoff.pdf` — (L) is (11) at every RK stage against the assembled $M^{-1}\mathcal{R}$, (R) is (11) once per step against the element's own $M_e^{-1}\mathcal{R}_e$. $c_f^{\max} = \sqrt{(\gamma p + |\boldsymbol{B}|^2)/\rho}$, the fast speed bounded over directions. ν_c: the residual coefficient before the $C_{\min}$ floor ν_{fl}. $T = p/\rho$. Unset keys take `src/io/mod_inputs.jl`'s defaults; environment overrides are taken unset. Notes *a*–*d*: next page.

Notes to the table

a The Brio–Wu deck sets $(C_1, C_2, C_{\min}) = \big(((N+1)/N)^2,\ 0.5\,(N+1)/N,\ 0.06\,(N+1)/N\big)$ on the kernel's $h_e/(N+1)$, i.e. $(1, 0.5, 0.06)$ on Dao & Nazarov's $\Delta_e = h_e/N$, the same at every N for a fixed number of DOFs. The paper varies $C_{\min}$ (`JEXPRESSO_BW_CMIN`).

b The Euler vortex uses the CompEuler kernel: $\mu_{\text{res}}|_e = C_1 \Delta_e^2\, n_\rho\, r_e$ and $\mu_{\max}|_e = C_2 \Delta_e \|\rho\|_{\infty,\Omega_e} \|\|\boldsymbol{u}\| + c\|_{\infty,\Omega_e}$ are dynamic (n_ρ: the density denominator); the momentum residual enters as one vector norm; heat conduction $\mathrm{P}\mu/(\gamma-1)$ with the artificial Prandtl number $\mathrm{P} =$ `:Pr` $= 0.1$ (default). The deck's `:dsgs_Prt => 0.7` is not read by this kernel.

c The paper's 160×70 run is `fluxEmergenceSon2025DSGS`. Its sibling `fluxEmergenceSon2025` (80×35, $\Delta t = 7.5 \cdot 10^{-3}$, $t_{\text{end}} = 54$, a case positivity floor as stage limiter) diffuses $\boldsymbol{q} - \boldsymbol{q}_e$ with ν (no ρ_e weight) and is otherwise the same.

d The deck's default is one point of the convergence figure; `JEXPRESSO_SV_NOP`, `_NELX` (`JEXPRESSO_EV_*` for the Euler vortex) set the others, with Δt from the rule.

Where the paper's text and the decks differ

These are the red entries of the table, with what each one changes.

1. **Sensor, (11).** The paper's (11) reads the assembled rate $M^{-1}\mathcal{R}$, which is the legacy sensor (L) when it is evaluated at every stage. The smooth vortices, Brio–Wu and the jet run (R): the element's own rate $M_e^{-1}\mathcal{R}_e$, once per step. Orszag–Tang, the rotor, the blast and the flux emergence run (L).
2. **Mass diffusion.** The paper sets $\lambda_1 = 0$ (mass untouched); every deck has `:μ[1] = 1`, so $\nu\nabla\rho$ is on, in all eight setups.
3. **Energy.** The paper's Table *split* has $\lambda_4 \bar{\rho}_e \mu|_e\, \gamma/((\gamma-1)\mathrm{Pr}_t)$ on ∇T with $\mathrm{Pr}_t = 0.7$. No deck uses that combination: the physical-form decks (the MHD vortex, Orszag–Tang) take Dao & Nazarov's $\bar{\rho}_e \nu/\mathrm{Pr}_t$ (`:dsgs_nazarov_energy`) with $\mathrm{Pr}_t = 1$, and the conserved-form decks diffuse E itself (the flux emergence, its thermal part only).
4. **Ohmic heating.** The paper's remark says the resistive term is not returned to the energy. The physical form (the MHD vortex, Orszag–Tang) adds the resistive work $\nabla \cdot (\eta \boldsymbol{B} \cdot \nabla \boldsymbol{B})$ to the energy, so the magnetic energy removed reappears as heat and the total energy is conserved.
5. **Conserved form.** The paper's stabilized system (stress, $\kappa_e \nabla T$, $\boldsymbol{\tau} \cdot \boldsymbol{u}$) is the physical form. Brio–Wu, the rotor, the blast, the jet and the flux emergence diffuse the conserved variables with one kinematic ν instead: no stress, no work terms.
6. **Normalization, (12) and (16).** (12) normalizes by the domain spread; the rotor, the blast, the jet and the flux emergence normalize by the element's own spread, floored at the element's scales. (16) has $\epsilon_{\text{rel}} = 10^{-3}$; every deck runs $\epsilon_{\text{rel}} = 1$, the floor being the full physical scale.
7. **Constants and length.** $C_{\min}$ is not in (12)–(14); Brio–Wu runs 0.06 and the flux emergence 0.03. Brio–Wu uses $\Delta_e = h_e/N$, not (15).
8. **Per stage or per step.** The paper's algorithm evaluates DynSGS once per right-hand-side call (every stage); the (R) decks compute $\mu|_e$ once per step.
9. **Not in the paper.** The start-up hold (2 steps; none in the blast and the jet), the fast-speed floors (blast, jet), the residual of ρe (blast) and the reference weight ρ_e (flux emergence).
10. **Discretization.** The rotor (paper 256^2 at $N = 4$; deck 128^2 at $N = 3$), the blast (paper 128^2, 256^2 at $N = 4$; deck 64^2 at $N = 3$) and Orszag–Tang (paper 128^2 to 512^2 in the figures and 32^2 with $\Delta t = 5 \cdot 10^{-4}$ in the text; deck 64^2, $\Delta t = 2 \cdot 10^{-4}$) were run on other meshes than the decks'. The flux-emergence figure is at $t = 51\,\tau_0$; the deck stops at 50.

Defaults that changed while the paper's runs were made

A run made before these dates used the old behaviour, whatever its deck says now.

- **2026-09-10** (`39362212`): the residual sensor (R) becomes the default and `:dsgs_sensor => "legacy"` keeps the old one. Before, every DynSGS run used (L).
- **2026-09-23** (`8a81279c`): the floors (16) go from $\epsilon_{\text{rel}} = 10^{-3}$ to $\epsilon_{\text{rel}} = 1$ (`:dsgs_rel`). Before, a nearly uniform field (the density of a vortex, B_x in a shock tube) had its residual divided by 10^{-3} of its scale.
- **2026-10-14** (`b956ccd` for `DSGS_MHD`, `e12fca8` for the CompEuler `DSGS`): with the residual sensor, $\mu|_e$ is computed once per step. Before, at every stage, where the residual stalls at $O(\Delta t)$ on smooth flow (`dsgsCutoff.pdf`).
- **Last change of each deck:** MHD vortex and Brio–Wu `b956ccd` (2026-10-14), Euler vortex `e12fca8` (2026-10-14), Orszag–Tang `6829a2c` (10-08), rotor `7bd7bc2` (10-09), blast `e4fcf14` (10-12), jet `2590be6` (10-12), flux emergence `c1d5b46` (10-13).